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APPLIED RESEARCH

ViT-Based Multi-Scale Classification Using Digital Signal Processing and Image Transformation

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ABSTRACT The existing classification of time-series data has difficulties that traditional methodologies struggle to address, such as complexity and dynamic variation. Difficulty with pattern recognition and long-term dependency modeling, high dimensionality and complex interactions between variables, and incompleteness of irregular intervals, missing values, and noise are the main causes for the degradation of model performance. Therefore, it is necessary to develop new classification methodologies to effectively process time-series data and make real-world applications. Accordingly, this study proposes ViT-based multi-scale classification using digital signal processing and image transformation. It comprises feature extraction through digital signal processing (DSP), image transformation, and vision transformer (ViT) based classification. In the DSP stage, a total of five features are extracted through sampling, quantization, and discrete fourier transform (DFT), which are sampling time, sampled signal, quantized signal, and magnitudes and phases extracted through DFT processing. Subsequently, the extracted multi-scale features are used to generate new images. Finally, based on the generated images, a ViT model is applied to make multi-class classification. This study confirms the superiority of the proposed approach by comparing traditional models with ViT and convolutional neural network (CNN) models. Particularly, by showing excellent classification performance even for the most challenging classes, it proves effective data processing in terms of data diversity. Ultimately, this study suggests a methodology for the analysis and classification of time-series data and shows that it has the potential to be applied to a wide range of data analysis problems.

INDEX TERMS Digital signal processing, image transformation, multi-class classification, multiscale, time series, vision transformer.

I. INTRODUCTION

The classification of time-series data is an important research topic in various domains [1]. However, the inherent complexity and dynamic variation of time-series data pose several challenges when traditional classification methodologies are used [2]. For instance, recognizing patterns over time or modeling long-term dependencies is a highly challenging task, which directly impacts classification accuracy. Additionally, time-series data often have high dimensionality and complex interactions between variables, so existing models have difficulties effectively handling and learning from the data [3]. Moreover, real-world time-series data may entail incompleteness of irregular intervals, missing values, and noise, which are the main contributors to model performance degradation [4]. Therefore, developing new classification methodologies that can effectively handle the complexity of time-series data and be applied to real-world problems in various domains emerges as a significant concern in both research and industry.

The approach of transforming time-series data into images offers advantages in overcoming these challenges. Through image transformation, it is possible to express visually the temporal patterns and characteristics of time-series data. This means that deep learning models, especially CNN or ViT models, which have demonstrated strong performance in

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image analysis, can also be applied to time-series data. Furthermore, the transformation into images allows for a more effective capture of multi-scale information and complex patterns and helps a model learn them. With the recent advancement of deep learning and DSP technologies, the paradigm of time-series data analysis is being re-established [5].

Therefore, this study proposes a new methodology aimed at overcoming the limitations of traditional time-series data classification and making the most use of the benefits of image transformation. The proposed methodology consists of three main steps. In the first step, DSP technique is applied to extract significant features from time-series data. In the second step, based on the extracted features, data are transformed into images. In the last step, the transformed images are classified through the ViT model. This process is aimed at effectively modeling the complex patterns and dynamic variations of time-series data, and achieving high classification accuracy. By providing the specific methodology of the new approach, experimental results, and insights gained thereof, this study comprehensively evaluates the impact of the convergence of deep learning and DSP technology on the field of time-series data analysis.

The contributions of the proposed method in this study are as follows:

- Proposing a new approach through the fusion of signal processing and image conversion. By combining DSP and image conversion techniques, we propose a new methodology to extract deeper features from time series data and visualize them effectively. This approach provides a new perspective on time series data analysis and explores applicability in a variety of fields.
- Development of a high-performance multi-class classification model using ViT. By applying the latest deep learning model, ViT, to data classification, higher accuracy, and efficiency are achieved compared to existing models. This represents a significant technological advance in recognizing and classifying complex patterns in time series data.
- Verification of the excellence of the methodology through performance comparison with various base models. Through comparison with traditional time series analysis models such as long short-term memory (LSTM), gated recurrent unit (GRU), and temporal convolutional network (TCN), the performance of this research methodology is objectively verified. This demonstrates the excellence of the proposed model and sets a new standard contributing to the field of time series data analysis.
- Provides detailed analysis of the model through performance optimization and hyperparameter adjustment. Through various hyperparameter settings and performance optimization experiments, model performance is further improved and in-depth analysis is provided. Through this, we increase understanding of the model and suggest guidelines for future research and application.

The organization of this paper is as follows. In Section II describes Basic time-series Classification and Advanced time-series Classification. In Section III discusses DSP, Image Transformation, and ViT-based Multi-Scale Classification. In Section IV presents the results and performance evaluation. In Section V describes the conclusion of this study.

II. RELATED WORK

Time-series data analysis goes beyond traditional techniques and is advanced into more sophisticated techniques capable of understanding and classifying even more complex and high-dimensional data structures [6]. In this section, review recent advanced time-series classification techniques and research trends and discuss how they relate to this study.

A. BASIC TIME SERIES CLASSIFICATION

Time-series data analysis has emerged as a significant research topic in various fields. Particularly, time-series classification is applied to a wide range of areas, such as pattern recognition, predictive modeling, and decision-making systems [7]. In this section, these researchers provide an overview of basic time-series classification methodology and introduce important theories and algorithms widely adopted in this field.

Time-series data is defined as a collection of data points observed in a time sequence. Given the data characteristics, it is important to understand the inherent temporal continuity and patterns. Early research in time-series classification mainly focused on statistical methodologies [8]. For instance, Alnaa and Ahiakpor [9] introduced the autoregressive integrated moving average (ARIMA) model for the temporal structure of time-series data. This model transforms non-stationary time-series data into stationary data and thereby supports the prediction of future values. With the development of machine learning, various algorithms have been applied to time-series classification. However, there are limitations in capturing nonlinear patterns and sensitivity to model parameter selection. Wagner et al. [10] proposed the Fuzzy DTW and Fuzzy BOSS model by applying the k-NN algorithm to a new time-series classifier. The proposed model effectively handles uncertain labels by applying fuzzy theory to existing DTW and BOSS algorithms. It shows robust performance even with noisy data and especially outperforms conventional methods in handling uncertain labels. However, the proposed method increases computational costs and faces difficult model interpretation due to the fuzzy labels and complex algorithm structure.

Aside from that, algorithms such as decision tree, support vector machine (SVM), and random forest are widely used in time-series classification. These algorithms learn the inherent patterns and structures of the data and use them to classify the data. Deng et al. [11] proposed a time-series Forest using a random forest algorithm. Based on the statistical features computed at various time intervals, it employs an ensemble of decision trees. This approach effectively classifies features

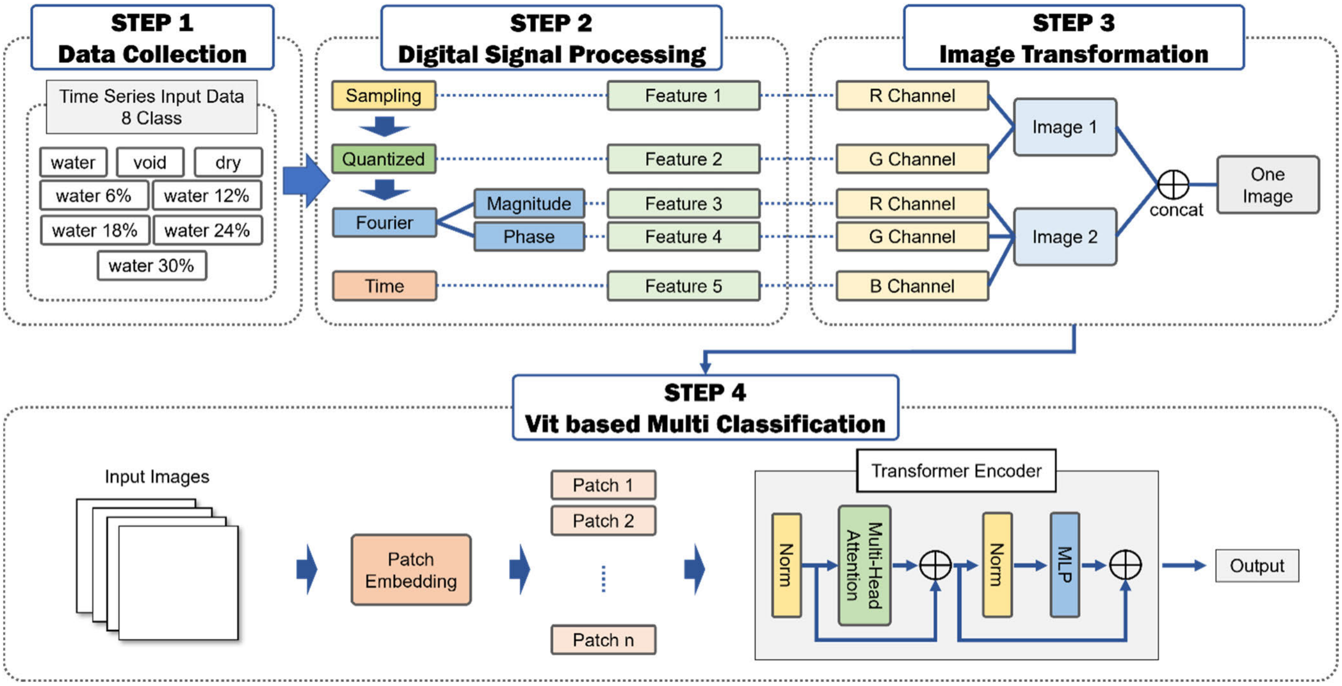

FIGURE 1. Process of ViT-Based multi-scale classification using digital signal processing and image transformation.

of time-series data and achieves high classification accuracy even with high-dimensional time-series data. However, considering diverse features and splitting criteria can make a model complex interpretation.

B. ADVANCED TIME SERIES CLASSIFICATION

Recently, deep learning techniques have been increasingly applied to time-series classification. Karim et al. [12] used LSTM to address time-series data classification. They proposed a novel model combining LSTM with fully convolutional networks. The proposed model makes it possible to achieve high classification performance and minimize data preprocessing. However, it has such limitations as an increased model size and complex model interpretation. Elsayed et al. [13] utilized a GRU for time-series analysis. GRU handles long-term dependencies in a similar way to LSTM but has a simpler structure which leads to higher computational efficiency. Therefore, it has lower model complexity than LSTM and allows for faster learning, effectively capturing important information in time-series data. However, compared to LSTM, GRU can have a limitation in modeling expressiveness, which results in low performance.

These days, such models as 1D CNNs and TCN demonstrate good performance when being applied to time-series classification. Koh et al. [14] used a TCN model to address time-series classification problems. TCN is based on 1D convolutional neural networks and was designed to effectively learn long-term dependencies in time-series data. It employs data processing and connection operations to deliver temporal characteristics of data to deep network layers. In this way, it is possible to effectively capture the temporal structure of time-series data and to improve classification performance. However, it causes high computational costs and a high potential for overfitting due to network complexity.

In addition to deep learning techniques, various ensemble methods and feature extraction techniques are being researched to effectively classify the nonlinear patterns of time-series data. For example, Fawaz et al. [15] revealed that ensembling various deep learning models can improve the performance of time-series classification. Moreover, research findings show that it is possible to further enhance the classification performance of models by applying appropriate feature extraction methods.

Based on these advanced time-series classification techniques, this study proposes a new approach by combining DSP and image transformation techniques. This approach is aimed at capturing more detailed and intricate data structures, compared to traditional time-series classification methods, and exploring the potential for effectively classifying them. Therefore, this study is expected to contribute to achieving technical advancements and providing a new perspective in the field of time-series classification.

III. METHODOLOGY

This study consists of a total of 4 stages. In the first stage, the data to be used in experiments are collected. In the second stage, digital signal processing is applied to the collected data. In the third stage, the features extracted through the DSP process are transformed into images. Lastly, in the fourth stage, multi-class classification is performed with the transformed images. Figure 1 shows the overall process of the methodology proposed in this paper.

TABLE 1. Experimental data measured by time domain reflectometry.

time(s) \ classes	dry	water	void	s6	s12	s18	s24	s30
0	0.0032009	0.002032	0.0006784	0.0039009	0.003194	0.0012639	0.0036097	0.0019064
0.01	0.00267	0.0023216	0.0005544	0.0005544	0.0026196	0.0009506	0.003519	0.0016406
0.02	0.0021272	0.0022626	0.000647	0.0022877	0.0018375	0.0005043	0.0032236	0.0012285
0.03	0.0017348	0.0018387	0.0008114	0.0014862	0.001087	0.0002151	0.0027168	0.0008543
…	…	…	…	…	…	…	…	…
81.90	1.0763296	0.9410283	1.1029843	0.9791709	0.9587644	0.958007	0.9471596	0.9390416
81.91	1.0762481	0.9414771	1.1025338	0.979541	0.9588419	0.954881	0.9472244	0.9393253

Firstly, significant features are extracted from time-series data through DSP process [16]. By sampling, quantization, and applying Fourier transform, multidimensional features are extracted through sampling, quantization, and Fourier transform. Based on them, multi-scale characteristics of the data are transformed into images [17], [18], [19]. With the images generated in the process, the complexity and multi-dimensionality of the data are effectively expressed with the use of RGB channels and a variety of color mapping techniques [20]. By applying multi-scale features to the images, it is possible to effectively address the potential issue of losing features in the course of the transformation of data into images [21].

Next, the multi-class classification approach using ViT is introduced [22]. Recently, ViT has demonstrated excellent performance in the field of image classification. It utilizes 232×5-sized images as inputs and effectively classifies various classes of the data used in the study. This study proves the excellent performance of ViT by comparing it with baseline models such as LSTM, GRU, and TCN [23], [24], [25]. Additionally, hyperparameter tuning experiments are conducted to optimize the performance of ViT and the impact of image size variation is analyzed [26].

Additionally, the reason why ViT models show higher performance than traditional CNN models is analyzed. Aside from that, the proposed methodology is compared with general image transformation methodologies such as spectrograms to clarify its distinctiveness in terms of performance [27].

A. DATA COLLECTION

The experimental data used in this study are the measured data of the reflection patterns of electromagnetic waves passing through concrete [28]. Experiments are conducted with the use of the created mold with 3m in length. The mold is divided into three sections of one meter each. Concrete is placed at both ends, and various mediums are used in the middle section for experimentation. As the medium changes, the reflection patterns of electromagnetic waves also change. Therefore, if other materials than concrete are present, it is possible to make a detection by catching out differences in reflection patterns. This enables the identification of defects in concrete structures, such as internal cavities, necking, and cracks.

The data generated from the experiments have a total of eight classes. These include 'void' for the empty air state, 'dry' for the state of filling with dry earth, 'water' for the state of filling with only water, and water with different moisture content percentages: 'water 6%(s6)', 'water 12%(s12)', 'water 18%(s18)', 'water 24%(s24)', and 'water 30%(s30)'.

The experiments include reflection coefficient values from about 0 to 82 seconds at an interval of 0.01 seconds. The reflection coefficient is the value of the reflection pattern of electromagnetic waves measured through time domain reflectometry (TDR) in an experimental setting. Each class consists of 2 columns and 8191 rows which represent time and reflection coefficient. Figure 2 shows the graphs of time and reflection coefficient for the eight classes.

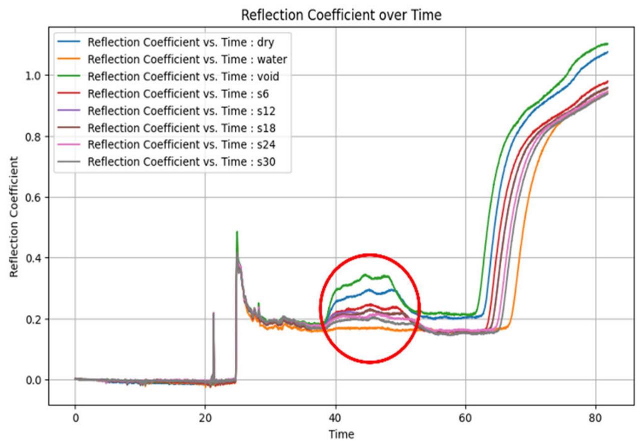

FIGURE 2. Time and reflection coefficient graphs for 8 classes.

In Figure 2, the horizontal axis represents Time, and the vertical axis represents Reflection Coefficient. At the ends of the graph, similar patterns are observed, but around the middle where a red circle is present, different patterns emerge. The green graph represents 'void', and the reflection coefficient of about 0.37 around 40 seconds. In the case of 'dry', the reflection coefficient is about 0.3 around the same seconds. 'Dry' and 'void' have similar patterns around about 50 seconds. In addition, the reflection coefficients of the purple graph 'water 12%' and the brown graph 'water 18%' are found to be similar.

B. FEATURE EXTRACTION USING DIGITAL SIGNAL PROCESSING

In this study, in the DSP stage, time-series data are processed into a suitable format for image transformation. Through the DSP, a variety of multi-scale information is captured from the time-series data, and the features to be applied to the image transformation are extracted. The DSP stage consists of three steps: sampling, quantization, and Fourier transform.

Algorithm 1 represents DSP, illustrating the process described in Section III-B. It depicts the processes from data loading to the creation of five features.

Algorithm 1 Digital Signal Processing

Input: input_path // Paths to csv files
Output: output_path //
preocedure DSP(csv_file)
  Load Data:
  dataframe ← pd.rcad_csv(csv_filc)
  time ← first column of *dataframe*
  signal ← second column of *dataframe*
  Sampling:
  sampled_df ← select data from *dataframe* matching specific condition
  sampled_time ← first column of *sampled_df*
  sampled_signal ← second column of *sampled_df*
  Quantization:
  quantized signal ← apply quantization function to (*sampled_signal,* desired levels)
  Perform DFT:
  dft result ← apply DFT to *(quantized signal)*
  magnitude ← calculate magnitude from *d* ftrresult ⁻
  phase ← calculate phase from *dft_result*
  return *sampled_time, sampled_signal, quantized_signal, magnitude, phase*
end preocedure

Figure 3 shows the results from the application of sampling to source data.

As shown in Figure 3, sampling is applied to each of the 8 classes. The orange point in Figure 3 represents the sampled point. Sampling is the process of converting a continuous-time signal into a discrete-time signal [17]. It is carried out as a way of measuring and recording continuous signals at a specific time interval. The primary purpose of sampling is to convert a signal into a digital format and thereby process it in computers or digital devices. In this study, sampling is conducted at an interval of 0.1 seconds. This makes it possible to extract key sampled data points from the time-series data. This process adjusts the temporal resolution of data, and these data points are used as the first feature. The next step is the quantization of the sampled data. Figure 4 shows the application of quantization to the sampled data.

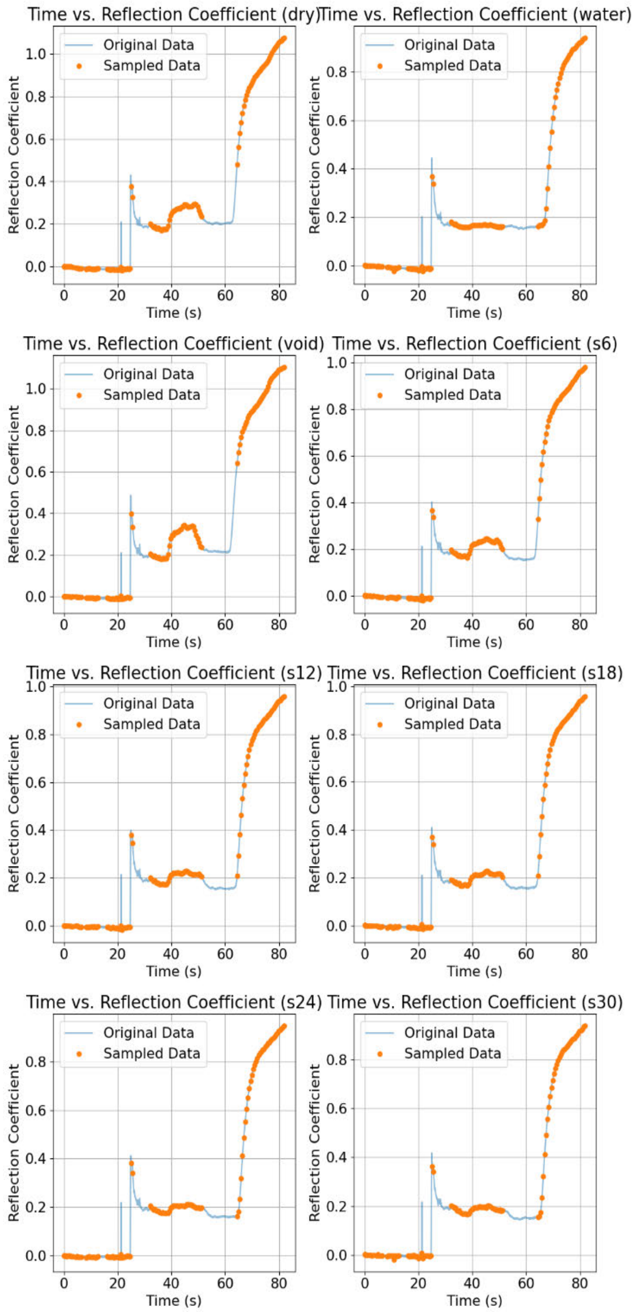

FIGURE 3. Results of applying sampling to source data.

As shown in Figure 4, quantization is applied to the sampled data. The orange point represents the sampled point, and the green point is the quantized point. Quantization is the process of limiting a sampled value to a specific resolution and converting the analog value of a signal into a digital value. In other words, it refers to the process of expressing a range of continuous values at a level of a finite number. In quantization, the value of each sample is rounded to the nearest standard value. A higher resolution of quantization allows for more precise expression of original signals, but it requires more storage space and processing power. In this study, a range from the minimum to maximum values is

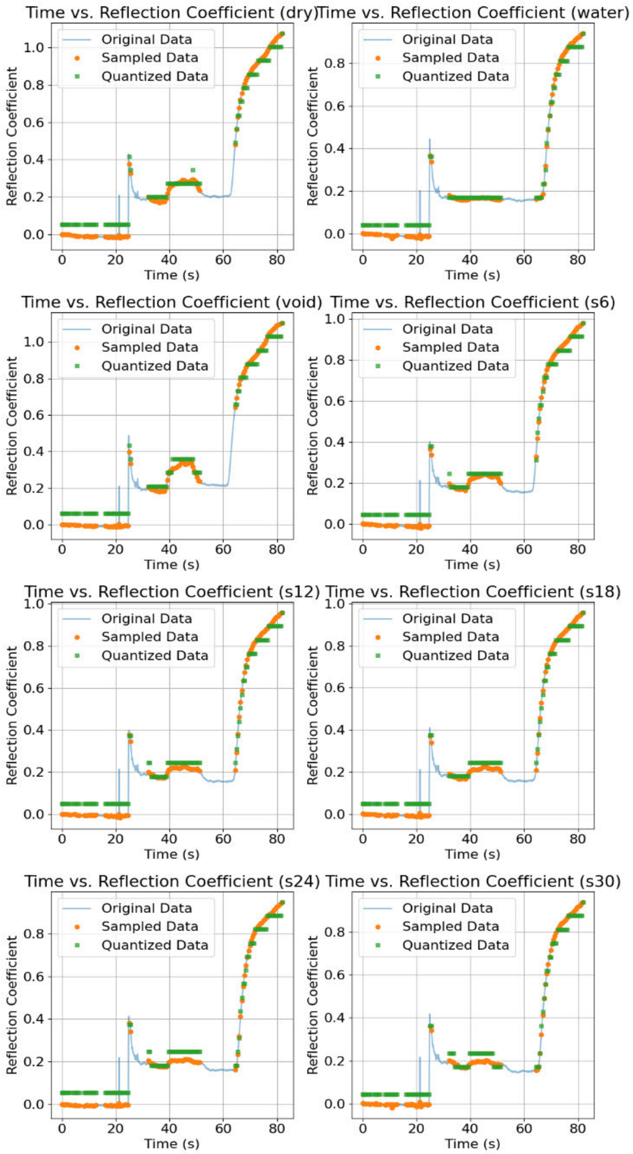

FIGURE 4. Results of applying quantization source data.

quantized in 16 levels. The quantized data point adjusts the size of the data and is used as the second feature. The next step is Fourier transform. Figure 5 shows the Fourier transform results for the quantized data.

As shown in Figure 5, DFT is applied to the quantized data. The blue solid line represents the Magnitude obtained through the DFT process, and the orange dotted line represents the Phase. Since a signal is converted into a discrete-time domain signal in the quantization process, DFT, rather than Fourier transform is used. The DFT is the process of converting a discrete-time domain signal into a frequency domain [19]. This makes it possible to analyze the frequency components of a signal. The two kinds of main information obtained through the transformation are Magnitude and Phase. Magnitude is the size of each frequency component, indicating how strong that frequency component is in the signal. Magnitude is a crucial factor in understanding

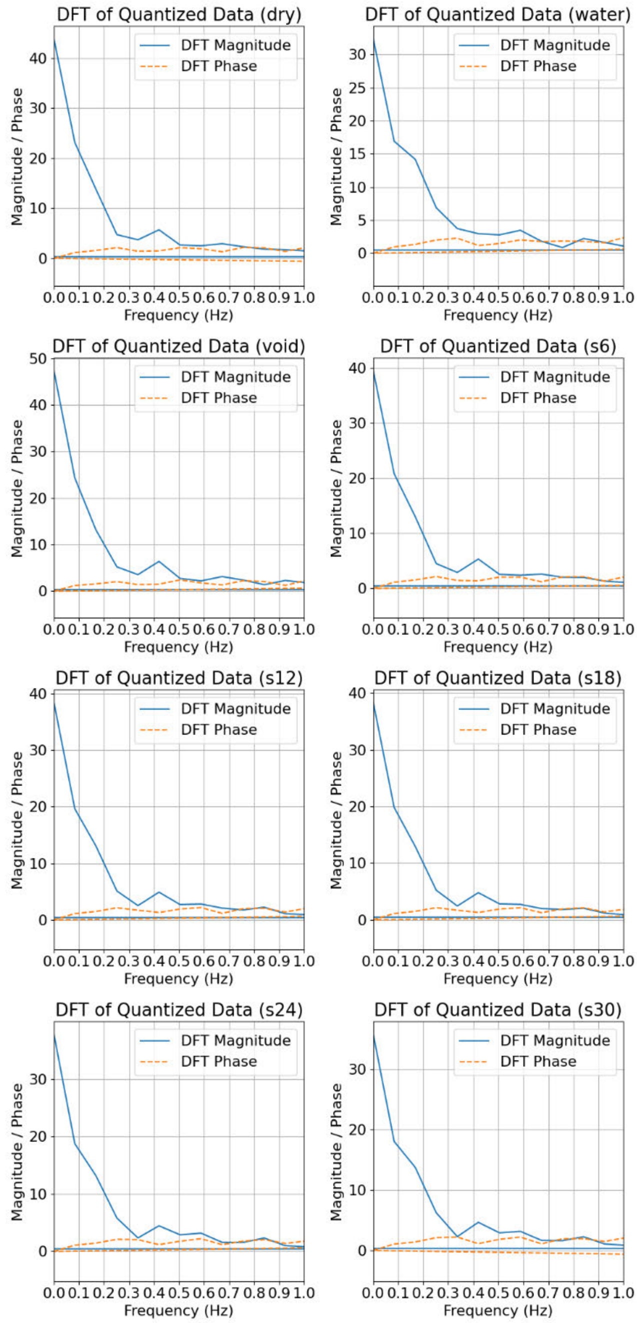

FIGURE 5. Results of applying discrete Fourier transform source data.

the energy distribution of a signal, helping identify which frequency components are dominant. Phase is the phase angle of each frequency component, providing information about the temporal structure and shape of a signal. Phase impacts a signal's shape and timing. Even with the same magnitude, having different phases can result in completely different signal shapes. The Magnitude and Phase extracted through DFT can be used to capture the complex composition and characteristics of signals, making it possible to analyze the basic structures and patterns of signals. In this study, Magnitude and Phase are used as the third and fourth features, respectively.

Lastly, the time information of time-series data is used as the fifth feature that reflects the temporal characteristics of data. This is crucial in capturing the dynamic features of time-series data. These extracted five features comprehensively reflect the multi-scale information of data. This helps to preserve the essential characteristics and patterns of data as much as possible, even if inevitable compression of information occurs during image transformation. Therefore, the DSP stage serves as the foundation for extracting important information from data and effectively converting them into images.

C. IMAGE TRANSFORMATION USING EXTRACTED FEATURES

This section describes the process of converting time-series data into images with the use of the five features extracted in Section III-B The purpose of this transformation is to preserve the multi-scale characteristics of time-series data and visually express the data. The transformation process involves creating two separate images and then combining them.

The process of generating the first image is as follows: The magnitude of the DFT, the phase of the DFT, and the value of time data are normalized in a range between 0 and 1. This normalization ensures that each feature can be appropriately mapped to the pixel value of the image channel. With the use of the normalized features, a 3-channel image is constructed. In this case, the R channel is set to the normalized value corresponding to the magnitude of DFT, the G channel to the normalized value corresponding to the phase of DFT, and the B channel to the normalized value corresponding to time data.

The process of generating the second image is as follows: The values of the sampled signal and the quantized signal are normalized in a range between 0 and 1. These two features are used to construct the first and second channels of the image. With the use of the normalized features, a 3-channel image is constructed. In this case, the R channel is set to the normalized value of the sampled signal, the G channel to the normalized value of the quantized signal, and the B channel to the combination or extension of these two values. Since this study lacks one feature, the B channel is created through the extension based on the information from the R and G channels.

The two images generated through the above two processes visually express each feature. These two images are concatenated horizontally to create a final image of size 5×232. Additionally, an image of size 224×224 is generated in line with the input of Vision Transformer model. Later, both the original and resized images are used to conduct performance evaluation to find a balance of the image size optimized for input specifications. These generated images contain a variety of multi-scale information of time-series data and aim to overcome the information compression that may occur during image transformation.

This image transformation method makes it possible to express the complexity and diversity of time-series data visually and effectively, and the transformed images will be used in the multi-class classification of ViT. This process demonstrates the possibility of maintaining the multi-scale characteristics of time-series data and achieving effective classification performance.

Figure 6 shows the appearance of the final image generated for the eight classes, and Figure 7 presents the resized image.

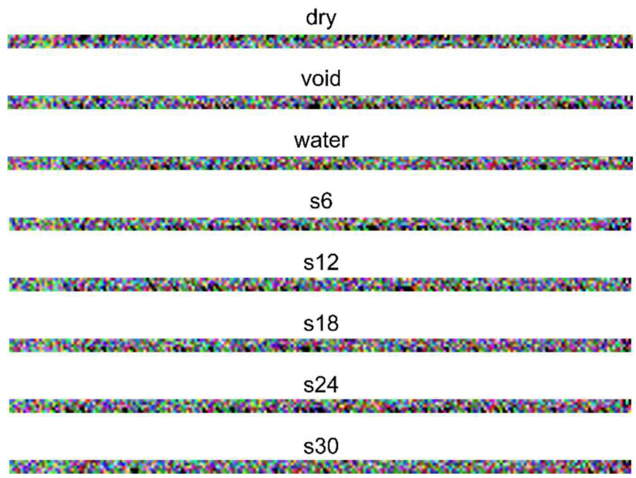

FIGURE 6. Image results were created from five features for eight classes.

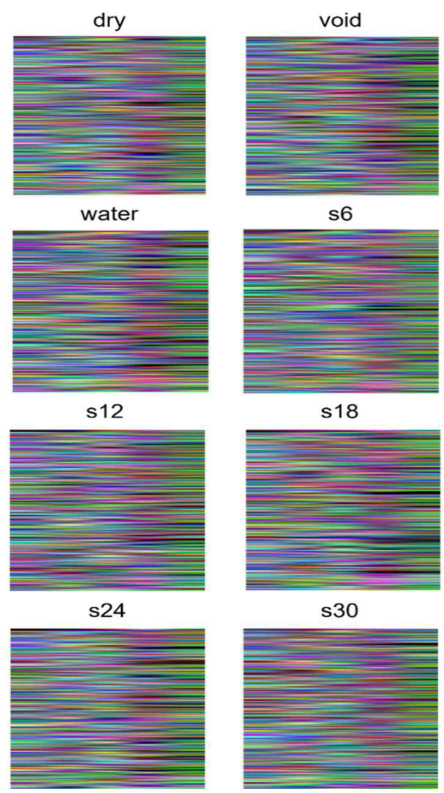

FIGURE 7. Resized image results were created from five features for eight classes.

Algorithm 2 shows the Image Transformation, depicting the process described in Section III-C and the process of generating images from the five features.

Algorithm 2 Image Transformation

Input: input // *sampled_time, sampled_signal, quantized_signal, magnitude, phase*
Output: output_path //
preocedure IMAGE TRANSFORMATION*(sampled_time, sampled_signal, quantized_signal, magnitude, phase)*
3 features to image:
norm_magnilude ← NORMALIZE(*magnilude*)
norm phase ← NORMALIZE(*phase*)
norm time ← NORMALIZE(*time*)
image 1 ← STACKCHANNELS*(norm_magnitude, norm_phase, norm_time)*
2 features to imags:
norm_sampled_signal ← NORMALIZE(*sampled_signal*)
norm quantized signal ← NORMALIZE(*quantized signal*)
dummy channel ← CREATEDUMMYCHANNEL
image2 ← STACKCHANNELS*(norm sampled signal, norm_quantized_signal, dummy_channel)*
Create composite image from features:
composite_image ← COMBINEIMAGESIDEBYSIDE*(image 1, image 2)*
return *composite_image*
end preocedure

By transforming time-series data into images, it is possible to effectively express the multidimensional features and temporal and spatial patterns of the data [29]. This helps deep learning models understand and learn better the complex structures of data. Furthermore, it is possible to integrate multi-scale information during image transformation. Such integration is good for capturing various frequencies and time-scale patterns of time-series data. For these reasons, time-series data are transformed into images [30].

D. ViT-BASED MULTI-SCALE CLASSIFICATION

This section describes the process of classifying the transformed images of time-series data using ViT model [22]. Compared to the traditional methods for analyzing time-series data or other deep learning-based image classification models, ViT model has shown higher accuracy and efficiency [31]. This indicates that ViT model is especially advantageous in transforming the complex features of time-series data into images and processing them. Moreover, ViT model provides deeper insights and predictive power than traditional approaches for analyzing time-series data [32]. The visualization of the time-series data obtained through image transformation enables ViT model to better understand and classify complex patterns [33]. Therefore, this study employs ViT model for the image transformation and classification of time-series data.

Figure 8 shows the structure of the ViT model used in this paper. ViT-Base model is used to classify the images of time-series data. The model maintains basic settings, and only an input size is adjusted. The input image size for the model is set to 232 × 5, corresponding to the size of the image generated in Section III-C. The patch size for processing the input images is set to 1. Consequently, the 232 × 5 image is divided into a total of 1,160 patches and put into the ViT model. In this paper, an input image is directly fed into the ViT model without any preprocessing, and thus no separate Feature Extractor is employed. The divided patches are passed through Linear Projection of Flattened Patches and then are put into the encoder of the transformer. The encoder of the transformer follows the structure of a conventional Transformer, consisting of 12 layers. After passing through the encoder, the patches go through MLP Head for final class classification.

The classification is conducted with the uses of both the original image transformed from time-series data (232 × 5) and the resized image (224 × 224) tailored to fit the input of ViT model. This allows for the evaluation of the impact of image size variation on classification performance. In the performance evaluation process, the ViT model is compared with other transformer-based vision models and traditional CNN models. This evaluation aims to assess how effective the ViT model is for image classification of time-series data. With the changes in the hyperparameters of the ViT model, multiple experiments are carried out to draw optimal performance. In this process, the performance

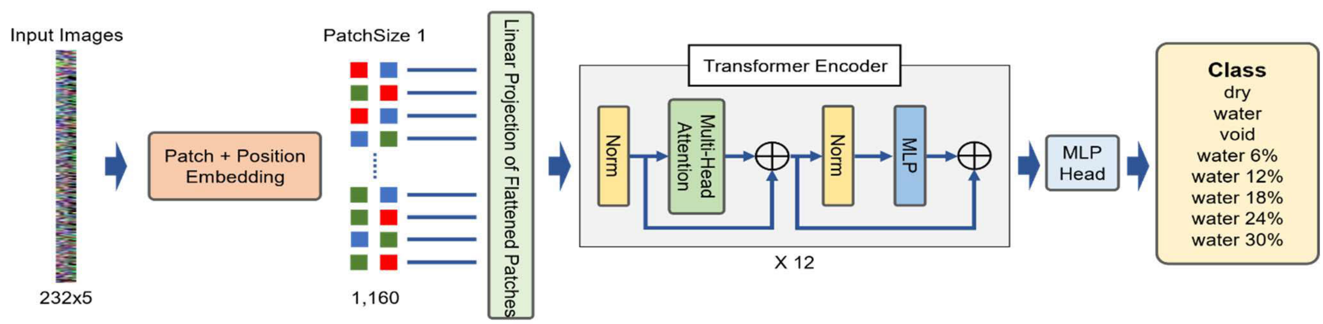

FIGURE 8. **Multi-class classification model structure based on ViT.**

changes of the model under various settings are observed and analyzed.

This study focuses on exploring the multi-class classification method of time-series data using the ViT model, and systematically evaluating the impacts of image size and model structure variation on classification performance. Therefore, it tries to look into the applicability and limitations of state-of-the-art deep learning models in time-series data analysis.

IV. EXPERIMENTAL RESULTS AND COMPARISONS

A. DATA AUGMENTATION AND PERFORMANCE EVALUATION METRICS FOR EXPERIMENTS

This section describes the experimental results and data augmentation. All experiments were conducted with two units of Intel®Xeon®Silver 2.40 GHz CPU, 128GB RAM, and NVIDIA RTX 3090 GPU. In this paper, noise injection is used as a data augmentation technique to increase the diversity of experimental data [34]. Data augmentation is applied to the experimental data (time-series data) acquired in Section III-A. Noise injection generates the transformed versions of data by adding random noise to the experimental data. The purpose of this process is to enhance a model's generalization ability by exposing it to various noises that may occur in real-world environments [34].

As a result, 30,000 augmented data samples per class are used as training data, and 9,000 augmented data per class as test data. In the training process, 10% of the training data are set aside as validation data and used. All experiments are conducted with the same augmented dataset, and the transformed images are also generated from the augmented data. This ensures consistent conditions for comparing and evaluating the experimental results.

Accuracy is an evaluation metric of experiments [35]. The Accuracy metric is used to measure how accurately a model predicts the classes of input images and is calculated as the ratio of correctly classified data to total data. Eq. (1) presents the calculation method for accuracy. True Positives (TP) are correctly identified positive cases, True Negatives (TN) are correctly identified negative cases, False Positives (FP) are negative cases incorrectly identified as positive, and False Negatives (FN) are positive cases incorrectly identified as negative.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

Additionally, Precision and Recall are used as evaluation metrics to evaluate the class-wise performance of traditional time-series data models and ViT. Precision represents the ratio of true positive items among the items classified as positive by the model. The Precision metric is used to measure how accurately a model predicts a positive class [36]. Recall represents the percentage of items correctly classified as positive by a model among actual positive items. The Recall metric is used to measure how well a model captures an actual positive class [37]. Eq. (2) and Eq. (3) represent the calculation methods for Precision and Recall, respectively.

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

ViT-MSC (Multi-Scale Classification), introduced in Section IV, refers to the ViT model which reflects the feature extraction and image transformation methods proposed in this paper. It is based on ViT-B, where Patch Size is set to 1 and image resizing is not applied.

B. COMPARISON WITH TRADITIONAL TIME SERIES MODELS

This section presents a comparative performance analysis between the ViT model and conventional time-series models, especially, LSTM, GRU, and TCN. The evaluation metric used for comparison is accuracy which represents the ratio of correctly predicted instances to total instances. The experiments include three existing time-series models, LSTM, GRU, and TCN, as well as the ViT model. The accuracy results for each model are presented below. Table 2 shows the comparison with existing time-series models.

TABLE 2. Comparison with traditional time series models.

Models	Accuracy
LSTM	88.07%
GRU	87.49%
TCN	62.41%
ViT-MSC(Ours)	**89.17%**

The ViT model used in this paper outperforms existing time-series models, achieving the highest accuracy of 0.8917. This indicates that the ViT model is excellent for capturing and classifying patterns of time-series data when the data are transformed into images. The performance advantage of the ViT model may stem from the following several factors: Firstly, the ViT model makes use of the advantages of rich feature expressions obtained through image transformation, making it possible to effectively capture complex patterns of data better than traditional models. Unlike traditional time-series models, the ViT model utilizes spatial relationships within the transformed images to provide additional contextual information which helps to enhance classification accuracy. The ViT model inherently has excellent scalability and flexibility so that it can handle various image sizes and complexity relatively easily.

The reason why GRU has slightly lower performance than LSTM is attributable to their different model structures. GRU has a simpler structure and fewer parameters than LSTM. For this reason, it has higher computational efficiency. However, compared to LSTM, GRU may have limitations in capturing detailed information when dealing with complex time-series data. Therefore, due to its simpler structure, GRU is speculated to have slightly lower performance than LSTM. The primary reasons why TCN model has relatively lower performance are the characteristics of its model structure and the complexity of time-series data. Although the TCN is designed to handle long-term dependencies, it is sensitive to specific hyperparameter settings. For this reason, the TCN may not adequately capture the various patterns and complexity of the given time-series data. Additionally, TCN's fixed dilation path may have limitations in effectively learning the diverse scales and patterns of the transformed image data.

In conclusion, the outstanding performance of the ViT model underscores the efficiency of the image transformation used for time-series data classification. This approach not only enhances model performance but also highlights the potential of vision-based models in the domain traditionally dominated by sequence-based models. The results of this study support the exploration of image-based models and transformation and present a promising method for future research on time-series analysis.

C. COMPARISON AMONG VISION TRANSFORMER MODELS

In this section, these researchers make a comparison between four ViT models: ViT-MSC (Ours), Swin-T [38], Swin-Tv2 [39], and PVT [40]. In a related experiment, resizing is applied to other models than the ViT model. The accuracy of each model is presented below. Table 3 shows the comparison between ViT models. Table 4 presents the key hyperparameter information used in the experiment.

TABLE 3. Comparison among vision transformer models.

Models	Accuracy
Swin-T	83.22%
Swin-Tv2	81.15%
PVT	85.97%
ViT-MSC(Ours)	**89.17%**

TABLE 4. Key hyperparameter information used in the experiment.

Models	Patch Size	Resize
Swin-T	4	Yes
Swin-Tv2	4	Yes
PVT	[4, 2, 2, 2]	Yes
ViT-MSC(Ours)	**1**	**No**

The ViT model has higher accuracy than different ViT models. That is mainly attributable to model flexibility and data adaptability. The ViT model can be used in line with the original data size of 232×5 without resizing. Since its Patch Size is set to 1, the model can capture detailed information of data. This is because the structure of ViT aligns well with the specific data structure in this study and because all parts of an image can be utilized effectively through precise patch size settings. It is speculated that other Vision Transformer models show slightly lower performance because they need resizing in terms of the characteristics of their model structures.

The reason why PVT outperforms Swin-T is that its pyramid-shaped structure is effective for capturing the features of multi-scales. PVT has a structure to hierarchically learn the features at different resolutions. The structure allows the PVT to effectively handle diverse patterns and scales of time-series data. This structural characteristic plays a key role in enabling PVT to achieve higher accuracy than Swin-T. It is estimated that the reason why Swin-T performs better than Swin-Tv2 is attributable to the fact that the Swin-T model has a more suitable nature for the image transformation of time-series data in this paper. Although Swin-Tv2 achieved more improvements and optimization structurally than Swin-T, these changes may have compromised the model's adaptability to a specific data structure. Particularly, in terms of the image transformation of time-series data, the structure of Swin-T might have been better suited to the data characteristics in this paper.

This comparative analysis reveals that the ViT model, compared to different ViT models, exhibited excellent performance in multi-class classification for the image transformation of time-series data. This indicates that the structural flexibility and data adaptability of the ViT model served as key roles for performance.

D. COMPARISON WITH CNN MODELS

In this section, for performance analysis, these researchers compare the ViT model with the latest CNN models such as ConvNeXt [41], ConvNeXtv2 [42], EfficientNet [43], and ResNet [44]. The comparison focuses on the accuracy achieved by each model when they classify the images transformed from time-series data. Table 5 presents the performance comparison with CNN models.

TABLE 5. Comparison with CNN models.

Models	Accuracy
ConvNeXt	84.69%
ConvNeXtv2	81.04%
EfficientNet	82.92%
ResNet	72.81%
ViT-MSC(Ours)	**89.17%**

Table 5 shows that the ViT model had better performance than recent CNN models. There are three main reasons why the ViT model outperforms the CNNs. Firstly, in the methodology used in this study, the transformation of time-series data into images generates images that are suitable for processing at the patch level. The ViT model splits an image into patches and independently processes them. This approach is good for capturing detailed information from the transformed images effectively. It enables the ViT model to utilize both local and global information effectively. Secondly, CNN models are known to have a significant inductive bias in handling spatial hierarchy and locality. Although this bias may be advantageous for certain types of images, it can also limit a model's adaptability. In contrast, the ViT model processes images in a patch sequence and shows a low level of inductive bias that does not impose strong spatial assumptions. This feature allows the ViT model to adapt better to the images transformed from time-series data that have no inherent inductive biases in natural images. Lastly, the images generated in the methodology proposed in this paper have the absence of traditional inductive biases such as continuity and locality which CNNs inherently exploit. As a result, the patch-based approach of the ViT provides a more suitable framework for capturing unique patterns and features of these transformed images.

The performance comparison emphasizes the effectiveness of the ViT model which outperforms traditional CNN models in terms of handling the images transformed from time-series data. ViT's patch-based processing and low inductive bias contribute to adapting to the unique characteristics of the transformed images and thereby achieve higher classification accuracy. This analysis suggests that the ViT model has potential in such areas as time-series data analysis beyond the field of natural image processing.

E. COMPARISON WITH SPECTROGRAM-BASED IMAGE TRANSFORMATION

In this section, these researchers compare the methodology used in this study with the most widely used spectrogram transformation method for transforming time-series data into images and analyzing them [45]. This comparison focuses on evaluating the performance of two approaches that are used to transform time-series data into visual forms. Table 6 presents the performance comparison between the spectrogram methodology and the methodology used in this study.

TABLE 6. Comparison with spectrogram-based image transformation.

Models	Accuracy
Spectrogram	12.73%
Ours	**89.17%**

Table 6 shows that the methodology used in this paper has higher performance than the spectrogram transformation method. The main reason why the methodology used in this study achieves higher accuracy than the spectrogram transformation method is that multi-scale information is effectively extracted from time-series data through DSP. The image transformation methodology used in this paper captures various scales and characteristics of time-series data through the DSP. This helps minimize information compression and loss that may occur during image transformation and preserves the complexity and diversity of data. Although the spectrogram transformation mainly focuses on visualizing the information in the frequency domain, the detailed information in the time domain may be lost in the process. Additionally, the spectrogram created with the time-series data used in this study has a size of 775×300. When the image is resized to 224×224 in line with the ViT model, more information is lost [46]. In contrast, the methodology used in this paper considers the information in both the time and frequency domains and preserves more information in the transformed images. In addition, it offers flexibility for the adaptation to various characteristics of time-series data. The DSP-based multi-scale information extraction makes it possible to capture data patterns in diverse scenarios and conditions and contributes to increasing accuracy in classification tasks.

This comparative analysis reveals that the methodology suggested in this study outperforms the traditional spectrogram transformation method in terms of image transformation and classification of time-series data. The suggested methodology transforms complex time-series data into visual forms more effectively and minimizes information loss that may occur in the transformation process. This indicates that the suggested methodology is an effective alternative for time-series data analysis and classification.

F. CLASS-WISE PERFORMANCE COMPARISON BETWEEN LSTM, GRU, TCN MODELS, AND ViT MODEL

In this section, these researchers conduct a performance comparison between the ViT model and LSTM, GRU, & TCN models regarding various classes of time-series data. In particular, the researchers analyze the accuracy of each model by class to evaluate how well the models classify specific classes of time-series data. Table 7 presents the class-by-class performance evaluation for the four models.

According to the performance analysis of LSTM, GRU, and TCN models, they show relatively low accuracy for the two most challenging classes, water 12% and water 18%. Additionally, they tend to have a bias of considering Class 4 and Class 5 as a single class. This indicates that these models have limitations in capturing complex patterns or subtle temporal variation. If time-series data of certain classes have intricate dynamic characteristics, traditional time-series processing models may have difficulty in adequately learning and classifying them. In contrast, the ViT model that reflects the methodology suggested in this paper demonstrates high classification accuracy even for the classes hard to find by conventional models. Furthermore, it consistently distinguishes between two classes, rather than having a bias of considering them as one class. That is because, in the suggested methodology of transforming time-series data into images, the ViT model is capable of effectively capturing and understanding multi-scale patterns and complex features of data. Particularly, the patch-based approach makes it possible to capture detailed information of data. That is a main factor contributing to the excellent performance of the ViT model compared to other models.

The ViT model and the image transformation methodology used in this paper offer a remarkable approach to time-series

TABLE 7. Class-wise performance comparison between LSTM, GRU, TCN models, and ViT-MSC model.

Classes	Accuracy							
	LSTM		GRU		TCN		**ViT-MSC(Ours)**	
	Precision	Recall	Precision	Recall	Precision	Recall	Precision	Recall
Class 0(dry)	1.00	1.00	1.00	1.00	0.99	0.99	**0.98**	**0.98**
Class 1(void)	1.00	1.00	1.00	1.00	0.99	0.99	**0.99**	**0.99**
Class 2(water)	1.00	1.00	1.00	1.00	0.38	0.19	**0.99**	**0.98**
Class 3(water 6%)	1.00	1.00	1.00	1.00	0.82	0.99	**0.97**	**0.98**
Class 4(water 12%)	0.70	0.08	0.57	0.74	0.52	0.02	**0.62**	**0.61**
Class 5(water 18%)	0.51	0.97	0.58	0.45	0.39	0.83	**0.63**	**0.65**
Class 6(water 24%)	1.00	1.00	0.92	0.89	0.38	0.50	**0.96**	**0.96**
Class 7(water 30%)	1.00	1.00	0.95	0.94	0.58	0.47	**0.96**	**0.96**

data analysis. Compared to traditional time-series processing models, the ViT model demonstrates the capability to classify various classes of complex time-series data more accurately. These findings indicate that the ViT model will be able to open up a new possibility for analyzing time-series data, especially hard-to-classify classes.

G. PERFORMANCE COMPARISON WITH VARIATIONS IN ViT HYPER-PARAMETERS

In this section, to evaluate the impact of hyperparameter variations in the ViT model on performance, these researchers conduct performance comparisons according to whether to apply resizing and a change in patch size. This comparison focuses on exploring the flexibility of the ViT model and its applicability in various scenarios. Figure 9 shows the performance comparison according to variations in ViT hyper-parameters.

In the case where resizing is applied, the default patch size is set to 16, and the model performance is evaluated. In the case where resizing is not applied, patch sizes are set to 1 and 2, and the model performance is evaluated. When the patch size is set to 1 without resizing, the model shows the highest performance. This indicates that the ViT model can better capture the intricate features and patterns of data when processing detailed information of images at a patch level. Particularly, for the images transformed from time-series data, smaller patch sizes appear advantageous for the model to learn subtle data variations more accurately.

Due to computational constraints, this study utilizes only the ViT-Base model for experimentation. If larger models like ViT-L are employed, there is a possibility of achieving higher accuracy. This is because larger models can capture data complexity more effectively through a greater number of parameters and deeper structures.

In this study, the performance comparison analysis based on hyperparameter variations reveals that the detailed settings of the ViT model impact image transformation and classification of time-series data. Particularly, it was found that patch size and the application of resizing are crucial factors influencing model performance. This finding provides

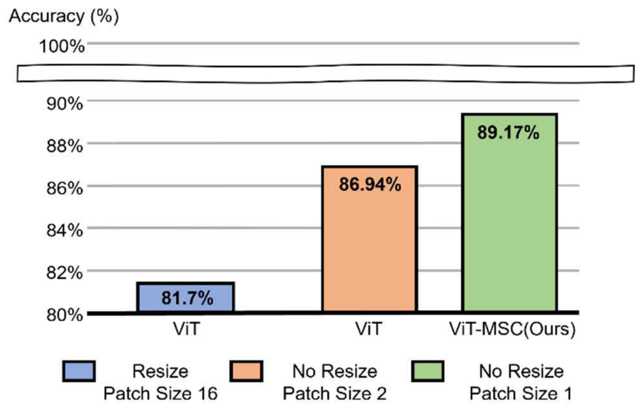

FIGURE 9. Performance comparison with variations in ViT Hyper-parameters.

guidelines for applying the ViT model to various data and tasks. In addition, this study confirms the potential for performance improvement with increasing model size. Therefore, it is expected that the scope and applicability of the ViT model will be extended further.

V. CONCLUSION

This study tried to explore a new approach to time-series data analysis by transforming time-series data into images and conducting classification with the use of ViT model. It was confirmed that the methodology developed in this study effectively transformed various time-series data into images. This indicates that the methodology is successfully applicable to reinterpretation of the intricate patterns and dynamic characteristics of time-series data in a visual form. The ViT model used in this paper demonstrated far better performance than time-series data models and CNN models. Particularly, when a patch size was set to 1, the ViT model had the best performance. This means that the detailed information generated during image transformation is significant. Additionally, the methodology of this study contributed to achieving high classification accuracy for all classes, including the most challenging ones. This suggests that transformed images can effectively reflect multi-scale information of time-series data. Additionally, the comparative analysis with spectrogram transformation method revealed that the image transformation methodology developed in this study has superiority and effectiveness. This indicates that the developed methodology can be used as a new approach to capture various features of time-series data.

The future research to be conducted has the following directions. The first one is to explore the construction of an end-to-end model by making parameters such as sampling and quantization in the DSP stage trainable. This will enable a model to automatically learn a transformation method more appropriate to data features to enhance performance further. The second one is to develop a method to vary time intervals in the sampling process to extract various features and integrate them into a single image. This aims to explore a method that can more richly reflect the multidimensional characteristics of time-series data. The last one is to develop a new model specialized for this methodology and then experiment with not only the ViT model but various structures and architectures. This aims to optimize the approach of the methodology and look into a new possibility of time-series data analysis. This study suggests a novel approach to time-series data analysis and will lay the foundation for future research and applications in the field.

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